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OC – CPU Performance Testing

Săptămâna 9

What's

PERFORMANCE

?

PERFORMANCE
WELNESS
TRANSFORMATION
NUTRITION
BEAUTY
NATURAL
EAT
WALK
REGENERATION
LIFESTYLE
EXERCISE
MOTIVATION
WEIGHT LOSS
ENERGY
FOCUS
VITAMIN
CALORIES
ASSESSMENT
FRUIT
ENERGY
GOALS
CARDIO
RUN
ACTIVE
TRAINING
POWER
MOVE
HEALTHY
SUCCESS
BODY
PERFORMANCE
ENDURANCE
MARATHON
FOOD
ACHIEVE
SPORT

performance
risk
plan
planning
analysis
customer
complex
data
process
opportunity
retention
impact
management
cost
evaluation
assessment
identification
probability
research
sales
project
organization
performance
treatment
strategy
resources
important
context
scope
evaluation

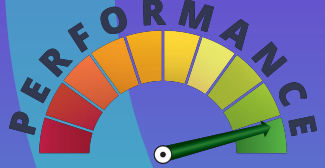
Cambridge Dictionary

What's ?

1. How well a person, machine, etc. does a piece of work or an activity
2. The action of entertaining other people by dancing, singing, acting, or playing music
3. How well someone does their job or their duties
4. Performance also refers to how well an activity or job is done
5. How successful an investment, company, etc. is and how much profit it makes
6. *[IT]* how well a computer, machine, etc. works

What's ?

1. (Despre sistemele tehnice, aparate, mașini) Care e susceptibil de un randament ridicat
2. Foarte competitiv
3. Rezultat (deosebit de bun) obținut de cineva într-o întrecere sportivă
4. Realizare deosebită într-un domeniu de activitate
5. *[IT]* Cel mai bun rezultat obținut de un sistem tehnic, de o mașină sau de un aparat

What's  ?

My opinion:

$$\frac{\text{Outcome (result)} * \text{importance}}{\text{Time + Energy (or what else we invest)}}$$

Types of testing

(some of them)

- **Testing is as important as any other part of the process!**
- Black-box vs. white-box testing
  - Advantages/disadvantages for each one of them?
- ***New!*** Grey-box testing
- Testing levels
  - Unit testing
  - Integration testing
  - System testing
  - Acceptance testing

Types of testing

(some of them)

- Smoke testing
- Regression testing
- Alpha and Beta testing
- Continuous testing → **Automated tests**
- Destructive testing → **Exceptions, failures, worst case scenarios**
- Performance testing

Performance testing

- Load testing → Scalability
- Stress testing → Extreme workloads
- Endurance testing → Expected load for a long period of time
- Spike testing → Sudden large spikes

Echipe

Puteți lucra (individual sau) în echipe

Despre app

Cât de complexă? Ce să includă?

Sugestii

Aplicații asemănătoare

Proiect

- Create a micro-benchmark app to test one or more functionalities of **CPU**

Suggestions of applications

CPU testing

- Fixed point arithmetics: conditional branching, array operations, recursivity, function calls, dhrystone etc.
- Floating point arithmetics: encryption, digits of Pi, convolution, image processing, fast fourier transform, trigonometry, whetstone etc.
- Threading, clock, caches & interprocess communication: recursivity, square root estimation, producer-consumer, hashing, cryptography, data access.
- Prime 95, Intel Burn Test, Orthos, Mandelmark, SuperPi, wPrime, others

Suggestions of applications

Digits of Pi

- Two categories of algorithms
- One can compute **consecutive digits based on the previous array of digits**
- The other approach may compute any digit in advance without having to go through every single digit; using this technique, researchers have recently found the 2*10^15-th digit of Pi

Any programming language

Standalone and portable

Windows 8+ OS mandatory, other OS optional

Easy to run

Click, double click, command line
No other input parameters

Provide the code

You will send me the source code
(you'll can use Git)

Idea

Originality or innovation of an existing idea

Application portability

Self explanatory

Software design

Code organization, algorithm correctness, efficient and unbiased implementation

Output presentation

User experience, ease of use, UI

THANK YOU FOR THE ATTENDANCE!

Ne vedem la test!